PS Analysis in Data Science: Problem Set 4

Problem 1. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x,y) = x^2 + y^2$.

(a) Compute $\nabla f(x,y)$.

(b) Evaluate it at $(1,2)$ and determine the local (affine) linear approximation of f in $(1,2)$.

(c) Plot f and ∇f in Julia.

Problem 2. Let $f : \mathbb{R}^3 \to \mathbb{R}$ be defined by $f(x,y,z) = xyz$.

(a) Compute $\nabla f(x,y,z)$.

(b) Evaluate it at $(1,1,1)$ and determine the local (affine) linear approximation of f in $(1,1,1)$.

(c) Plot $(x,y) \mapsto f(x,y,1)$ in Julia.

Problem 3. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x,y) = x^2 y + y^2$.

(a) Compute $\operatorname{grad} f(x,y)$.

(b) Determine the local (affine) linear approximation of f at $(1,1)$.

Problem 4. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x,y) = \frac{x}{x^2+y^2+1}$.

(a) Plot f in Julia.

(b) Guess its local extrema from your plot.

Problem 5. Let $f : \mathbb{R}^2 \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be twice continuously differentiable such that $f(x,y) = g(x+y)$.

(a) Show that the Taylor polynomial of g of order 1 centered at 0 and evaluated at $t = x+y$ coincides with the local (affine) linear approximation of f in $(0,0)$.

(b) The Hessian matrix of f is

$$H_f(x,y) = \begin{pmatrix} \partial_x \partial_x f(x,y) & \partial_x \partial_y f(x,y) \\ \partial_y \partial_x f(x,y) & \partial_y \partial_y f(x,y) \end{pmatrix}.$$

Consider $g(t) = e^t$ and show that the Taylor polynomial of g of order 2 centered at 0 and evaluated at $t = x+y$ coincides with

$$F(x,y) = f(0,0) + \operatorname{grad} f(0,0) \begin{pmatrix} x \\ y \end{pmatrix} + \frac{1}{2}(x,y) H_f(0,0) \begin{pmatrix} x \\ y \end{pmatrix}.$$